

LECTURE 15 BINARY RESPONSE MODELS

This lecture discusses models in which the dependent variable takes only two values, usually zero or one. For example, Y_i can measure whether or not individual i participates in the labor force, with $Y_i = 1$ (yes) and $Y_i = 0$ (no). The linear regression model may not be appropriate in such cases.

Bernoulli trials

The econometrician observes $\{Y_i : i = 1, \dots, n\}$, where the Y_i are iid random variables. The distribution of Y_i is given by

$$Y_i = \begin{cases} 1 & \text{with probability } \theta, \\ 0 & \text{with probability } 1 - \theta. \end{cases}$$

This is the Bernoulli(θ) distribution, and

$$\begin{aligned} E[Y_i] &= \theta \\ &= \Pr(Y_i = 1). \end{aligned} \tag{1}$$

The PMF of Y_i is

$$p(y_i, \theta) = \theta^{y_i} (1 - \theta)^{1-y_i} \text{ for } y_i = 0, 1.$$

Thus, the log-likelihood function is given by

$$\begin{aligned} \log L_n(\theta) &= \left(n^{-1} \sum_{i=1}^n Y_i \right) \log \theta + \left(1 - n^{-1} \sum_{i=1}^n Y_i \right) \log (1 - \theta) \\ &= \bar{Y}_n \log \theta + (1 - \bar{Y}_n) \log (1 - \theta), \end{aligned}$$

where $\bar{Y}_n = n^{-1} \sum_{i=1}^n Y_i$. The ML estimator of θ is the sample mean:

$$\hat{\theta}_n^{ML} = \bar{Y}_n.$$

Computing the second derivative,

$$\frac{d^2 \log p(Y_i, \theta)}{d\theta^2} = -\frac{Y_i}{\theta^2} - \frac{1 - Y_i}{(1 - \theta)^2}.$$

Equation (1) implies that

$$\begin{aligned} E \left[\frac{d^2 \log p(Y_i, \theta)}{d\theta^2} \right] &= -\frac{1}{\theta} - \frac{1}{1 - \theta} \\ &= -\frac{1}{\theta(1 - \theta)}. \end{aligned}$$

Thus, in this model, the information (scalar) is given by

$$I(\theta) = \frac{1}{\theta(1 - \theta)},$$

and the results in Lecture 14 imply that

$$\begin{aligned} \bar{Y}_n &\rightarrow_p \theta, \\ n^{1/2} (\bar{Y}_n - \theta) &\rightarrow_d N(0, \theta(1 - \theta)). \end{aligned}$$

We can estimate the asymptotic variance consistently by $\bar{Y}_n (1 - \bar{Y}_n)$. A $1 - \alpha$ asymptotic confidence interval for θ can be constructed as follows:

$$\left[\bar{Y}_n \pm z_{1-\alpha/2} \sqrt{\frac{\bar{Y}_n (1 - \bar{Y}_n)}{n}} \right].$$

The Bernoulli trials model is univariate. We now extend it to the case where the probability of Y_i taking the value 1 is a function of some exogenous explanatory variables.

Linear probability model

Suppose that the econometrician observes $\{(Y_i, X_i) : i = 1, \dots, n\}$, where the Y_i are binary $\{0, 1\}$ variables, and the X_i are k -vectors of exogenous variables that explain $\Pr(Y_i = 1 \mid X_i)$. Let us assume that

$$\Pr(Y_i = 1 \mid X_i) = F(X_i^\top \beta),$$

for some function F . Unlike the Bernoulli trials model, the probability of Y_i taking the value one or zero is not constant and depends on some observable characteristics X_i .

Under the linear regression model,

$$E[Y_i \mid X_i] = X_i^\top \beta,$$

then

$$F(X_i^\top \beta) = X_i^\top \beta.$$

That is, the probability of $Y_i = 1$ is a linear function of X_i . Such an assumption is a good starting point for the analysis, but it suffers from a serious flaw. If some of the components of X_i are continuous, $X_i^\top \beta$ cannot be restricted to the zero-one interval, and the model may yield probabilities outside $[0, 1]$. To avoid such predictions, we require F to map into $[0, 1]$, which forces F to be nonlinear. Any CDF is a natural choice for F .

Despite this limitation, the LPM remains useful in practice for estimating average partial effects, particularly when the main interest is in a treatment coefficient. Under correct specification of the conditional expectation, OLS provides a consistent estimator of the average marginal effect without requiring a distributional assumption on the errors.

Probit and logit models

It is convenient to introduce a latent (unobservable) variable Y_i^* , which can be interpreted as the net benefit or unobserved utility that determines the agent's binary decision. The usual linear regression model holds for Y_i^* :

$$Y_i^* = X_i^\top \beta + U_i.$$

We assume further that $\{(Y_i^*, X_i) : i = 1, \dots, n\}$ are iid, and that

$$Y_i = \begin{cases} 1 & \text{if } Y_i^* > 0, \\ 0 & \text{if } Y_i^* \leq 0. \end{cases}$$

Let F be the conditional CDF of U_i given X_i :

$$F(u \mid X_i) = \Pr(U_i \leq u \mid X_i).$$

Then,

$$\begin{aligned} \Pr(Y_i = 1 \mid X_i) &= \Pr(Y_i^* > 0 \mid X_i) \\ &= \Pr(X_i^\top \beta + U_i > 0 \mid X_i) \\ &= \Pr(U_i > -X_i^\top \beta \mid X_i) \\ &= 1 - F(-X_i^\top \beta \mid X_i). \end{aligned}$$

Assume further that the distribution of U_i does not depend on X_i , so $F(u)$ replaces $F(u | X_i)$. The usual assumption in this framework is that F is symmetric around zero, i.e.,

$$F(u) = 1 - F(-u).$$

Under this assumption,

$$\Pr(Y_i = 1 | X_i) = F(X_i^\top \beta).$$

The common choices for F are

- Probit model: F is the standard normal CDF, denoted by Φ .
- Logit model: F is the logistic CDF, denoted by Λ :

$$\begin{aligned} \Lambda(X_i^\top \beta) &= \frac{\exp(X_i^\top \beta)}{\exp(X_i^\top \beta) + 1} \\ &= \frac{1}{1 + \exp(-X_i^\top \beta)}. \end{aligned}$$

For either choice of F , the model is estimated by maximum likelihood. The conditional PMF of Y_i given $X_i = x_i$ takes the same form as in the Bernoulli model, with θ replaced by $F(x_i^\top \beta)$:

$$p(y_i, \beta | x_i) = F(x_i^\top \beta)^{y_i} (1 - F(x_i^\top \beta))^{1-y_i} \text{ for } y_i = 0, 1.$$

Thus, the log-likelihood is given by

$$\log L_n(\beta) = n^{-1} \sum_{i=1}^n Y_i \log F(X_i^\top \beta) + n^{-1} \sum_{i=1}^n (1 - Y_i) \log (1 - F(X_i^\top \beta)).$$

The first-order condition for $\hat{\beta}_n^{ML}$ is

$$\begin{aligned} 0 &= \frac{\partial \log L_n(\hat{\beta}_n^{ML})}{\partial \beta} \\ &= n^{-1} \sum_{i=1}^n Y_i \frac{\partial F(X_i^\top \hat{\beta}_n^{ML}) / \partial \beta}{F(X_i^\top \hat{\beta}_n^{ML})} - n^{-1} \sum_{i=1}^n (1 - Y_i) \frac{\partial F(X_i^\top \hat{\beta}_n^{ML}) / \partial \beta}{1 - F(X_i^\top \hat{\beta}_n^{ML})} \\ &= n^{-1} \sum_{i=1}^n Y_i \frac{f(X_i^\top \hat{\beta}_n^{ML})}{F(X_i^\top \hat{\beta}_n^{ML})} X_i - n^{-1} \sum_{i=1}^n (1 - Y_i) \frac{f(X_i^\top \hat{\beta}_n^{ML})}{1 - F(X_i^\top \hat{\beta}_n^{ML})} X_i, \end{aligned}$$

where f is the density corresponding to F :

$$f(u) = \frac{dF(u)}{du}.$$

No closed-form expression exists for $\hat{\beta}_n^{ML}$; it must be computed numerically. Standard statistical software produces the ML estimates and asymptotic standard errors for both models.

For the logit model, the derivative of Λ satisfies

$$\frac{d\Lambda(u)}{du} = \Lambda(u) (1 - \Lambda(u)). \quad (2)$$

Therefore, for the logit model,

$$\begin{aligned}\frac{f(X_i^\top \beta)}{F(X_i^\top \beta)} &= 1 - \Lambda(X_i^\top \beta), \\ \frac{f(X_i^\top \beta)}{1 - F(X_i^\top \beta)} &= \Lambda(X_i^\top \beta).\end{aligned}$$

The first-order condition simplifies to

$$\begin{aligned}\frac{\partial \log L_n(\hat{\beta}_n^{ML})}{\partial \beta} &= n^{-1} \sum_{i=1}^n (Y_i - \Lambda(X_i^\top \hat{\beta}_n^{ML})) X_i \\ &= 0.\end{aligned}$$

Similarly,

$$\frac{\partial \log p(y_i, \beta | x_i)}{\partial \beta} = (y_i - \Lambda(x_i^\top \beta)) x_i,$$

and

$$\frac{\partial^2 \log p(y_i, \beta | x_i)}{\partial \beta \partial \beta^\top} = -\Lambda(x_i^\top \beta) (1 - \Lambda(x_i^\top \beta)) x_i x_i^\top.$$

Thus, the information matrix is given by

$$\begin{aligned}I(\beta) &= -E \left[\frac{\partial^2 \log p(Y_i, \beta | X_i)}{\partial \beta \partial \beta^\top} \right] \\ &= E [\Lambda(X_i^\top \beta) (1 - \Lambda(X_i^\top \beta)) X_i X_i^\top].\end{aligned}$$

Hence, for the logit model, $\hat{\beta}_n^{ML} \rightarrow_p \beta$, and

$$n^{1/2} (\hat{\beta}_n^{ML} - \beta) \rightarrow_d N \left(0, (E [\Lambda(X_i^\top \beta) (1 - \Lambda(X_i^\top \beta)) X_i X_i^\top])^{-1} \right).$$

The asymptotic variance-covariance matrix of $\hat{\beta}_n^{ML}$ can be estimated consistently by

$$\left(n^{-1} \sum_{i=1}^n \Lambda(X_i^\top \hat{\beta}_n^{ML}) (1 - \Lambda(X_i^\top \hat{\beta}_n^{ML})) X_i X_i^\top \right)^{-1}.$$

Marginal effects

In the linear regression model, the marginal effect of X_i on $E[Y_i | X_i]$ is β . In nonlinear binary choice models such as probit or logit,

$$\begin{aligned}\frac{\partial E[Y_i | X_i]}{\partial X_i} &= \frac{\partial \Pr(Y_i = 1 | X_i)}{\partial X_i} \\ &= \frac{\partial F(X_i^\top \beta)}{\partial X_i} \\ &= f(X_i^\top \beta) \beta,\end{aligned}$$

and

$$\begin{aligned}\frac{\partial \Pr(Y_i = 0 | X_i)}{\partial X_i} &= \frac{\partial (1 - \Pr(Y_i = 1 | X_i))}{\partial X_i} \\ &= -f(X_i^\top \beta) \beta.\end{aligned}$$

Thus, in the case of the probit model, the marginal effect of X_i on $\Pr(Y_i = 1 \mid X_i)$ is given by

$$\phi(X_i^\top \beta) \beta,$$

where $\phi(u) = d\Phi(u)/du$ is the standard normal density. In the case of logit, equation (2) implies that $\partial \Pr(Y_i = 1 \mid X_i) / \partial X_i$ is given by

$$\Lambda(X_i^\top \beta) (1 - \Lambda(X_i^\top \beta)) \beta.$$

The marginal effects can be estimated by replacing the unknown coefficients β with their ML estimators. For a fixed $X_i = x$,

$$\frac{\partial \Pr(\widehat{Y_i = 1} \mid X_i = x)}{\partial X_i} = f(x^\top \hat{\beta}_n^{ML}) \hat{\beta}_n^{ML}.$$

By Slutsky's theorem and the consistency of $\hat{\beta}_n^{ML}$,

$$\begin{aligned} \frac{\partial \Pr(\widehat{Y_i = 1} \mid X_i)}{\partial X_i} &= f(x^\top \hat{\beta}_n^{ML}) \hat{\beta}_n^{ML} \\ &\rightarrow_p f(x^\top \beta) \beta \\ &= \frac{\partial \Pr(Y_i = 1 \mid x)}{\partial X_i}, \end{aligned}$$

provided that f is continuous.

The asymptotic distribution of $f(x^\top \hat{\beta}_n^{ML}) \hat{\beta}_n^{ML}$ can be obtained using the delta method:

$$\begin{aligned} n^{1/2} \left(f(x^\top \hat{\beta}_n^{ML}) \hat{\beta}_n^{ML} - f(x^\top \beta) \beta \right) &\rightarrow_d N \left(0, \frac{\partial (f(x^\top \beta) \beta)}{\partial \beta^\top} I^{-1}(\beta) \frac{\partial (f(x^\top \beta) \beta^\top)}{\partial \beta} \right) \\ &= N \left(0, (f'(x^\top \beta) \beta x^\top + f(x^\top \beta) I_k) I^{-1}(\beta) (f'(x^\top \beta) \beta x^\top + f(x^\top \beta) I_k)^\top \right). \end{aligned}$$